Shovan Sahu

Kolkata, West Bengal, India | shovansahu000@gmail.com | +91 9832432155
linkedin.com/in/shovan-sahu | github.com/shovan-sahu

Summary

Backend Engineer with production experience designing scalable microservices, high-performance REST APIs and asynchronous data pipelines using Python (Django, FastAPI), Node.js, and Bun. Skilled in optimizing hybrid retrieval architectures, implementing Model Context Protocol (MCP) servers, containerization, and distributed system design. Focused on driving system reliability, reducing API latency, and executing clean data orchestration at scale.

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript
Runtimes & Frameworks: Django, FastAPI, Node.js, Bun, Express.js
Databases, Search & Queues: MongoDB, Redis, Elasticsearch, Qdrant, SQL, Bull.js
Infrastructure & DevTools: Docker, Docker Compose, Git, GitHub, Postman, MCP (Model Context Protocol)
Core Concepts: Microservices Architecture, REST API Design, RAG Pipelines, Async Background Processing, Event-Driven Workflows, Vector Embeddings, ETL Pipelines

Experience

Oggangs Pvt. Ltd. Feb 2025 – Oct 2025
Backend Developer Intern *Bangalore, India (Remote)*

- Developed a high-throughput Django microservice that engineered an internal data enrichment and ETL pipeline, extracting and cleaning raw text data with a **95% processing success rate**.
- Programmed APIs within an internal AI-driven calling agent microservice to extract structured information from conversations using LLMs, auto-generating clean product records and cutting manual data entry overhead by **60%**.
- Built and optimized core production REST APIs using Node.js (Express.js), handling secure user authentication and business logic for over **50,000+ active monthly requests**.
- Designed and deployed asynchronous background job pipelines utilizing Bull.js and Redis, offloading heavy data scraping workflows and cutting core API response latency by **42%**.
- Configured Elasticsearch clusters for full-text search and faceted filtering, achieving **sub-100ms retrieval speeds** across datasets exceeding 2M+ unstructured documents.
- Formulated optimized MongoDB indexing strategies for distributed, high-write database workloads and containerized services using Docker to guarantee uniform local and staging environments.

Projects

InsightsHub | *Python (FastAPI), TypeScript (Node.js), Qdrant, Groq, Docker* | insightshub.in 2026

- Built a decoupled FastAPI service owning the vector storage layer in Qdrant and high-speed inference pipelines via Groq API, boosting context retrieval accuracy to **88%** through optimized text chunking strategies.
- Architected a distributed, production-ready RAG microservices platform using Node.js with TypeScript for a strongly-typed, maintainable, and performant execution environment.
- Integrated **ElasticLake** (S3-compatible object storage) for scalable asset storage and implemented **Razorpay** subscription plans with webhook verification and payment lifecycle management.
- Containerized the complete application stack with Docker Compose, separating concerns to allow **100% independent service deployability** and streamlined horizontal scaling.

SocialLoop | *Express.js, MongoDB, Socket.io, Cloudinary* | socialloop.onrender.com 2025

- Developed a full-stack social media backend application using complex MongoDB aggregation pipelines to serve dynamic, optimized user feeds.
- Engineered a real-time messaging architecture using Socket.io, configuring optimized concurrent connection handling to minimize database read bottlenecks and ensure low-latency data packet delivery.

Education

Swami Vivekananda University Kolkata, West Bengal
Bachelor of Technology in Computer Science and Engineering *Graduation: 2025*

Certifications

Backend Developer Internship Certificate – Oggangs Pvt. Ltd. (2025)